CAUSAL INFERENCE & SYNTHETIC CONTROL ENGINE FOR ALGORITHMIC PRICING

Final Project Report

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This report documents an illustrative analysis using a synthetic retail-pricing dataset. Results should not be interpreted as verified real-world causal effects.

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1. Abstract

This project studies how a change in product price may be associated with a change in sales. It applies Difference-in-Differences (DiD), statistical tests and a Synthetic Control approach to a synthetic retail dataset. The treatment product is Earphones, whose listed price is changed from ■500 to ■540 on 15 May 2025. USB Cable is used as a comparison product and its price remains ■200. The calculated DiD estimate is approximately –1.901 units per day. The synthetic-control gap change is approximately –0.0646 units per day. These estimates differ and must be interpreted cautiously because the dataset is synthetic, the parallel-trends condition is not perfectly satisfied, and the synthetic donor series were generated using information related to the treatment series.

2. Introduction

Pricing decisions affect customer demand, revenue and business performance. A business may increase or decrease a product price to improve revenue, but a simple before-and-after comparison cannot establish that the price change caused a sales change. Other factors such as seasonality, promotions, product availability and customer preferences may also affect sales. Causal-inference methods provide structured ways to compare outcomes and make assumptions explicit.

3. Problem Statement

The project addresses the following question: after the Earphones price increase from ■500 to ■540, how did average daily units sold change relative to a comparison product and a constructed synthetic comparison? The analysis also asks whether the observed group differences are statistically detectable and how closely the synthetic series matches the treatment series before the intervention.

4. Objectives

- Prepare and inspect a structured pricing and sales dataset.
- Identify treatment and comparison groups and define the intervention date.
- Calculate before-and-after group averages.
- Estimate a Difference-in-Differences effect.
- Apply Welch’s t-test and Mann–Whitney U test.
- Evaluate a Synthetic Control comparison using error and gap metrics.
- Present findings, recommendations and limitations clearly.

5. Scope of the Project

The work focuses on a single illustrative price intervention, one treatment product and one comparison product, supported by a synthetic-control exercise. It covers data preparation, exploratory analysis, descriptive comparisons, statistical testing, model evaluation and reporting. It does not estimate a production-ready pricing model or prove how actual customers would respond in a live market.

6. Tools and Technologies

Python: data processing and analysis.
Google Colab / Jupyter Notebook: execution environment and documentation.
Pandas and NumPy: data handling and numerical calculations.
Matplotlib: charts and visual summaries.
SciPy: statistical hypothesis tests.

GitHub: project version control and sharing.
PDF: final written report format.

7. Dataset Description

The project dataset is synthetic and contains approximately 4,000 rows and 10 columns. The columns are Date, Product, Category, Price, Units_Sold, Price_Change, Treatment_Group, Control_Group, Post_Period and Revenue. The dates span 1 January 2024 to 22 June 2029. The intervention date used in the analysis is 15 May 2025. Earphones are the treatment product and USB Cable is the comparison product. The Earphones price is represented as increasing from ■500 to ■540, while USB Cable remains at ■200.

8. Data Preparation and Exploratory Analysis

The workflow includes loading the CSV file into a Pandas DataFrame, checking column names and data types, parsing the Date field, checking missing values and duplicates, validating numeric columns, and confirming the treatment, control and post-intervention indicators. Exploratory analysis summarizes sales and revenue by product, category and time period. The dataset should be rechecked in the submitted notebook to confirm the exact cleaning operations and row counts reported by the executed file.

9. Methodology

9.1 Difference-in-Differences (DiD)

DiD compares the before-to-after change in the treatment group with the before-to-after change in the comparison group. The estimate is:
DiD = (Treatment after – **Treatment before)** – **(Control after** – **Control before).**

9.2 Statistical testing

Welch's t-test compares group means without assuming equal variances. Mann–Whitney U is a rank-based test that compares distributions. Small p-values indicate evidence against the relevant null hypothesis; they do not, by themselves, prove causation.

9.3 Synthetic Control

A weighted combination of donor products is used as a synthetic comparison for the treated product. The pre-intervention fit is evaluated using RMSE. The difference between observed and synthetic outcomes is summarized before and after the intervention. A change in the gap is calculated as post-gap minus pre-gap.

10. Results and Interpretation

The treatment group average decreased from 24.49 units per day before the intervention to 22.54 after it, a change of –1.95. The control group average changed from 39.64 to 39.59, a change of approximately –0.0487. The resulting DiD estimate is approximately –1.9013 units per day.

For Earphones, Welch's t-test p-value is approximately 3.80 × 10■^{3}■ and the Mann–Whitney p-value is approximately 6.88 × 10■^{3}■. For USB Cable, the corresponding p-values are approximately 0.8143 and 0.9962. These tests indicate a detectable before/after difference for the treatment product in this synthetic dataset, but not for the control product. They do not establish that price alone caused the difference.

The Synthetic Control pre-intervention RMSE is approximately 0.7333 units per day and full-period MAE is

approximately 0.6163. The average pre-intervention gap is approximately 0.0486 and the post-intervention gap is approximately −0.0160. The synthetic gap change is therefore approximately −0.0646 units per day. This is much smaller than the DiD estimate, so the two approaches do not tell the same magnitude story.

11. Graphical Analysis

The following figures visualize the confirmed summary metrics recorded during the project. They are summary charts, not reconstructed daily time-series plots.

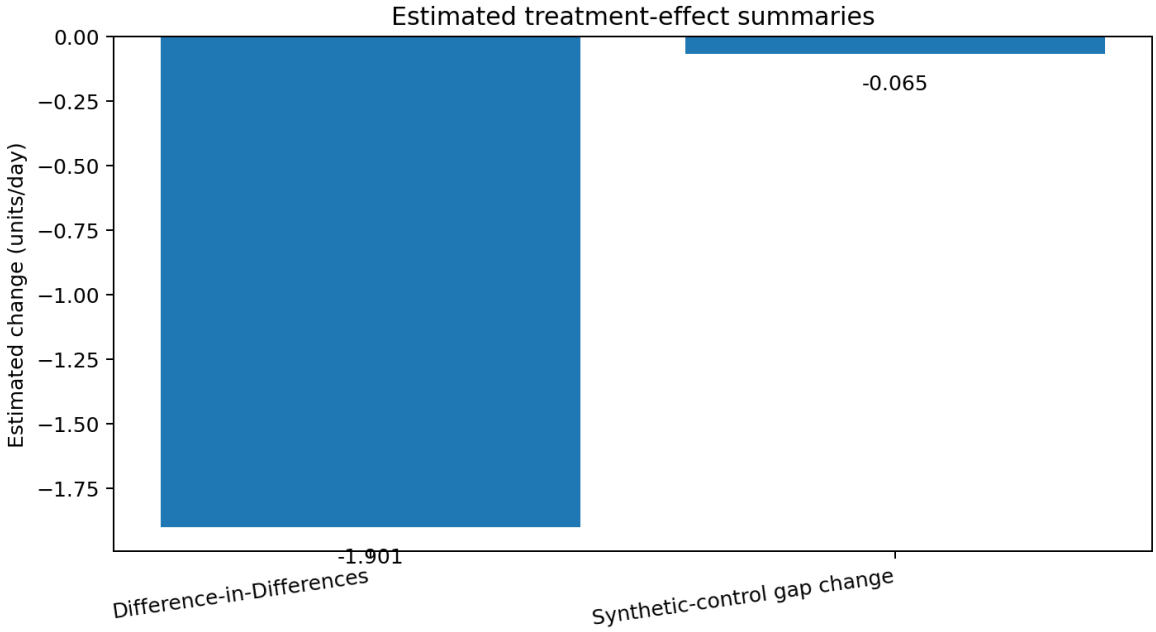


Figure 1. Group averages before and after the intervention. Earphones average daily sales decreased from 24.49 to 22.54. USB Cable averages changed from 39.64 to 39.59.

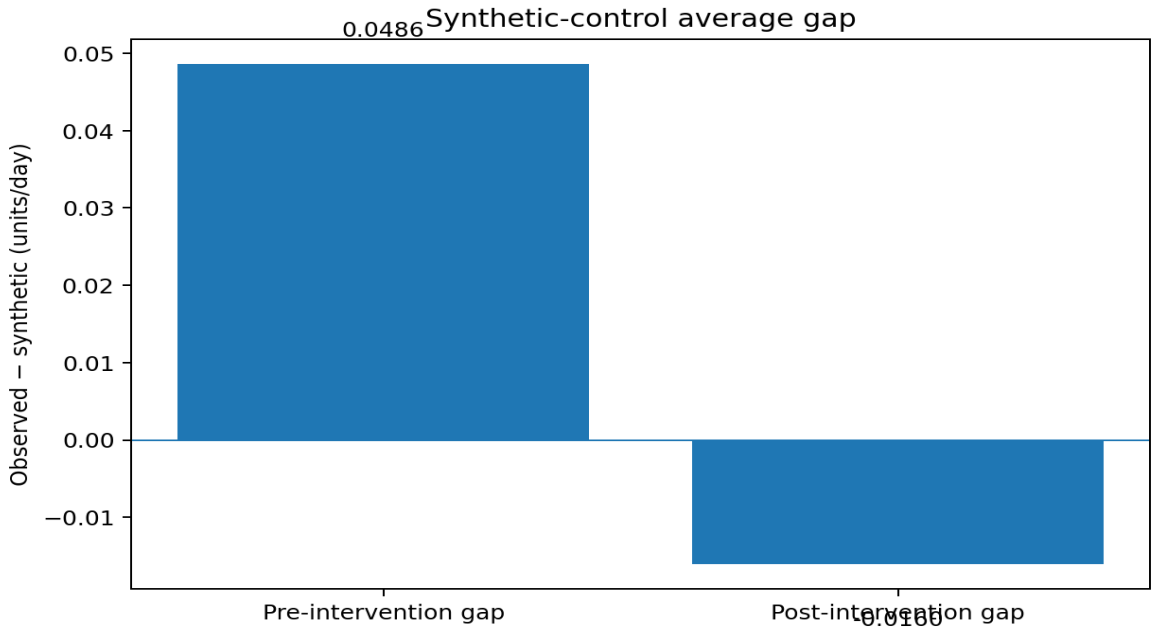


Figure 2. Estimated treatment-effect summaries. The DiD estimate is –1.9013 units/day, while the synthetic-control gap change is –0.0646 units/day. These estimates use different comparison constructions and should not be treated as interchangeable.

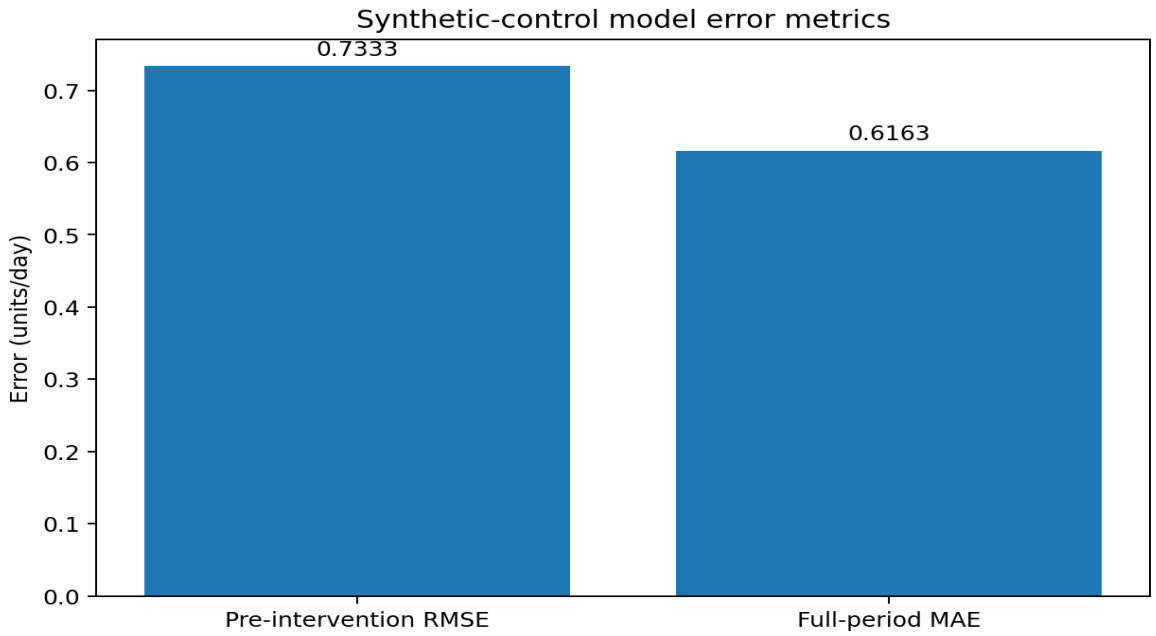


Figure 3. Synthetic-control average gap. The observed-minus-synthetic gap moved from 0.0486 before the intervention to –0.0160 after it.

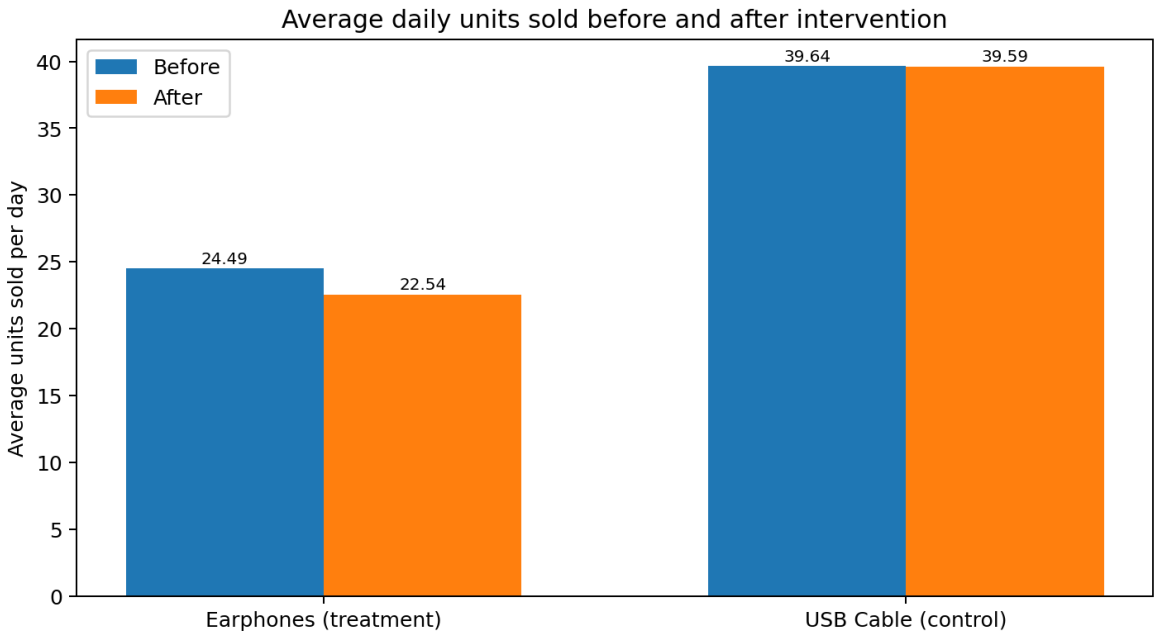


Figure 4. Synthetic-control error metrics. Pre-intervention RMSE is 0.7333 and full-period MAE is 0.6163 units/day.

12. Business Recommendations

• Treat price changes as experiments: where practical, use randomized or carefully matched comparison groups.
• Monitor units sold, revenue, profit margin, returns and customer response together; unit sales alone do not determine whether a price change is beneficial.
• Record promotions, stock-outs, advertising, holidays and competitor actions so these factors can be considered in the analysis.
• Validate pre-intervention trends before relying on DiD.
• Use independent donor series and test alternative donor combinations before relying on Synthetic Control.
• Avoid making operational pricing decisions from this synthetic demonstration alone.

13. Limitations

1. The dataset is synthetic, not a real retailer's transaction data.
2. Parallel trends were not perfectly satisfied, weakening the DiD interpretation.
3. The synthetic-control donor series were generated using the treatment Earphones series plus noise. They are not independent donor observations, which can make the synthetic-control fit and estimated gap unreliable.
4. The DiD and Synthetic Control estimates differ substantially.
5. Statistical significance does not remove bias from confounding factors or design limitations.
6. The analysis focuses on units sold and does not establish the effect on profit or long-term customer behavior.

14. Future Enhancements

The project can be improved by using real transaction-level data, adding more independent control products, checking pre-trends with visual and statistical diagnostics, estimating confidence intervals, conducting placebo intervention tests, testing alternative donor pools, and analyzing revenue and profit margin alongside units sold. A dashboard can also be developed to let users select products, intervention dates and comparison groups.

15. Conclusion

This project demonstrates a workflow for examining a pricing intervention using descriptive statistics, Difference-in-Differences, hypothesis tests and Synthetic Control. In the synthetic dataset, the DiD estimate is approximately −1.9013 units per day and the synthetic-control gap change is approximately −0.0646 units per day. The difference between these estimates, imperfect parallel trends and dependent synthetic donor construction mean that the results are illustrative rather than conclusive. The main outcome is a documented analytical workflow and an understanding of the assumptions that must be checked before using causal methods for business decisions.

16. References

1. Pandas documentation: https://pandas.pydata.org/docs/
2. SciPy documentation: https://docs.scipy.org/doc/scipy/
3. Matplotlib documentation: https://matplotlib.org/stable/
4. Difference-in-Differences and Synthetic Control concepts as studied and implemented in the project notebook.
5. Project dataset and analysis notebook maintained in the project GitHub repository.